

TY B.Sc. DS Sem VI After Oct 4

Time: 2½ Hours

Total Marks: 75

N. B.: (1) **All** questions are **compulsory**.(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.(3) Answers to the **same question** must be **written together**.(4) Numbers to the **right** indicate **marks**.(5) Draw **neat labeled diagrams** wherever **necessary**.(6) Use of **Non-programmable** calculators is **allowed**.**1. Attempt any three of the following:****15**

- Define Business Analytics and explain its necessity in data analytics. Provide examples of its real-world applications.
- Explain the measures of central tendency (mean, median, and mode) and discuss how they help in analyzing business data.
- How do data visualization techniques help in understanding relationships between different variables? Explain with examples.
- What are outliers and missing values in a dataset? Explain their impact on business analytics and strategies to handle them.
- Discuss the concept of datasets, variables, and observations with reference to business analytics. How are they essential in data-driven decision-making?
- How do graphical, algebraic, and spreadsheet models contribute to business decision-making? Provide suitable examples.

2. Attempt any three of the following:**15**

- Explain the probability distribution of a single random variable. How does it help in decision-making?
- Discuss the role of business intelligence tools in data analytics. Provide examples of commonly used tools and their applications.
- How does decision-making under uncertainty differ from decision-making under risk? Explain with suitable examples.
- What is the importance of probability distributions in business analytics? Explain with examples of normal, binomial, and Poisson distributions.
- Explain the concept of expected value and its significance in decision-making. How does it help in evaluating business strategies?
- Discuss the applications of Bayes' Theorem in business analytics. Provide an example to illustrate its practical use.

3. Attempt any three of the following:**15**

- Define sampling and discuss its importance in business analytics. How does it help in making inferences about a population based on sample data?
- Explain the central limit theorem and its significance in sampling distributions. Provide an example to illustrate its impact on business analytics.
- What are the different types of sampling methods used in business analytics? Discuss their advantages and limitations with examples.
- Discuss the factors that influence the width of a confidence interval, including sample size, confidence level, and population variability.
- Explain Type I and Type II errors in hypothesis testing. How do they impact decision-making in business analytics?
- Explain the concept of p-value in hypothesis testing. How does it help in determining the significance of a statistical test in business decision-making?

4. Attempt any three of the following:**15**

- a Define regression analysis and explain the process of fitting a straight line in simple linear regression. What method is commonly used for this purpose?
- b Explain different methods of forecasting. How do they help businesses in decision-making?
- c What is multicollinearity in regression analysis? How does it impact the interpretation of regression coefficients?
- d Discuss the significance of the random walk model in time series forecasting. Provide an example of its application in business analytics.
- e What are the key steps in building a time series forecasting model? Explain with suitable examples.
- f Explain the different methods used to evaluate the accuracy of a forecasting model. Why is accuracy assessment crucial in predictive analytics?

5. Attempt any three of the following:**15**

- a. Define optimization modeling and explain its significance in business analytics. How does it aid in decision-making?
- b. Explain the process of sensitivity analysis in optimization modeling. Why is it essential for evaluating the robustness of optimal solutions?
- c. What are the key differences between linear programming and integer programming? Provide examples to illustrate their applications.
- d. Discuss the various steps involved in the simulation modeling process. Why are model validation and verification crucial?
- e. What is two-way ANOVA? Explain its applications in business analytics with an example.
- f. Explain the key applications of simulation modeling in business analytics. How does it help in making data-driven decisions under uncertainty?

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- What methods can be used to evaluate a batsman's performance across different venues?
- How can a bowler's effectiveness be assessed using moving averages and economy rates?
- How can we compare the relative performance of multiple batsmen?
- How can Sachin Tendulkar's performance be analyzed with basic statistics?
- What statistical models can be used to forecast a batsman's future runs?
- What is Sachin Tendulkar's impact on matches won and lost?

2. Attempt any three of the following:**15**

- What are the functionalities of the five tabs in the *Sixer* app?
- What visualizations can be generated using *cricketr* for analyzing T20 cricket performances?
- How can a batsman's performance be demonstrated at home and overseas?
- How can a plot of 4s, 6s, and the scoring rate in ODIs be generated for batsman analysis?
- How does *cricketr* facilitate the analysis of all-rounders in cricket? Provide an example.
- Provide a detailed overview of the *cricketr* R package.

3. Attempt any three of the following:**15**

- How can the relative bowling performances of top ODI bowlers be measured using *cricketr*?
- What insights does the "Clash of the Titans" analysis provide about the greatest Test batsmen of all time?
- How can a complete career summary report of a player be generated using *cricketr* templates?
- Provide a brief analysis of the relative performances of international batsmen against Australia in Australia.
- What visualizations does *cricketr* offer for analyzing batsmen across different decades?
- How can the performance of a batsman against a host country in its home conditions be analyzed?

4. Attempt any three of the following:**15**

- a. How does *cricpy* facilitate the prediction of batting performance in ODIs?
- b. What visualizations can be generated for bowlers in T20 matches using *cricpy*?
- c. How can the contributions of all-rounders in T20 matches be analyzed using *cricpy*?
- d. How does *cricpy* assist in the comparative analysis of batsmen across different decades?
- e. How can player data be retrieved using the `getPlayerData()` function in *cricpy*?
- f. Write a short note on *cricpy*, a Python package for analyzing cricketers' performances.

5. Attempt any three of the following:**15**

- a. How can *cricpy* be used to compare average runs against top bowling teams across different decades?
- b. How does *cricpy* handle historical data for cross-era comparisons of cricket legends?
- c. Write a short note on analyzing the top three wicket-takers using *cricpy*.
- d. How does *cricpy* analyze a batsman's performance across different T20 leagues?
- e. How can a future runs forecast plot and a relative mean strike rate plot be generated using *cricpy*?
- f. How can the future bowling averages of players like Kapil Dev, Anil Kumble, and Bhagwat Chandrasekhar be predicted using machine learning in *Octave*?

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- a. What is Data Governance, and why is it important for organizations?
- b. Discuss real-world examples of Data Governance and its impact on business success.
- c. How does Data Governance contribute to data security, compliance, and business value?
- d. What are the challenges and best practices in implementing Data Governance in an organization?
- e. How can organizations leverage Big Data to improve decision-making and business outcomes?
- f. Explain the key components of an effective Data Governance framework.

2. Attempt any three of the following:**15**

- a. What are the key ingredients of Data Governance tools?
- b. What is the role of an Enterprise Dictionary in Data Governance?
- c. What are the phases of a Data Life Cycle?
- d. How can organizations apply governance over the Data Life Cycle?
- e. What is Data Lineage, and why is it important in Data Governance?
- f. How does Change Management impact Data Governance?

3. Attempt any three of the following:**15**

- a. What are the key priorities and outcomes of Data Governance in an organization?
- b. What are the main roles and responsibilities in a Data Governance framework?
- c. What is the significance of Data Governance leadership groups?
- d. What are the key steps in designing and implementing a Data Governance program?
- e. What is Data Quality, and why is it important for organizations?
- f. What are the main roles and responsibilities in a Data Governance framework?

4. Attempt any three of the following:**15**

- a. What are the key aspects of planning data protection in an organization?
- b. How is data protection implemented in the cloud, and what are its challenges?
- c. What is Identity and Access Management (IAM), and how does it enhance data security?
- d. What is data exfiltration, and what measures can organizations take to prevent it?
- e. What is monitoring in data governance, and why is it important?
- f. What are the key criteria and processes involved in an effective monitoring system?

5. Attempt any three of the following:**15**

- a. What is a data culture, and why is it important for organizations?
 - b. How does Data Governance benefit businesses?
 - c. Why are training and communication essential in Data Governance?
 - d. How does Data Governance interact with legal and security requirements?
 - e. What are the key steps in handling data-related incidents?
 - f. What are the major challenges and risks in Data Governance, and how can organizations respond to them?
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15

- Define and explain the Internet of Things with 2 examples.
- "Any sufficiently advanced technology is indistinguishable from magic". Discuss.
- List and explain the roles of people making IoT
- Discuss the issue of Privacy in Internet of Things.
- What are TCP and UDP ports? Explain with examples.
- "Data available through IOT device belongs to public or company which implements the IOT device".

2. Attempt any three of the following:

15

- Explain microcontrollers and system-on-chips with respect to embedded computing
- Describe the difficulties encountered during the transition from a prototype to mass production?
- How can one tap into the community for promoting IOT devices? Explain
- Discuss the trade-offs between cost versus ease of prototyping
- How is development done for Arduino? Explain.
- "Open source has a competitive advantage". Discuss

3. Attempt any three of the following:

15

- Explain the sketch, iterate and explore process in prototyping
- What are laser cutters? Explain the main features to consider while choosing a laser cutter.
- What is CNC Milling? Explain.
- Explain the following terms with respect to APIs: (a) Scraping, (b) Mashing up APIs.
- What is comet? Explain.
- What are non-digital methods and materials commonly used in prototyping?

4. Attempt any three of the following:

15

- a What are the concerns regarding performance and battery life while writing code for embedded systems?
- b Write a short note on long tail of Internet.
- c Explain different types of libraries for embedded systems which works with limited memory
- d Discuss the business model canvas for Internet of Things.
- e Explain the following business models:
 - i. Provide infrastructure
 - ii. Take a percentage
 - iii. Be a Key Resource
- f What is venture capital? How can one exit?

5. Attempt any three of the following:

15

- a. Explain the steps for manufacturing PCBs
 - b Discuss different environmental issues in Internet of Things.
 - c Write a short note on mass-producing the case and other fixtures.
 - d Explain in detail the process of designing kits.
 - e Discuss the issues in scaling up the software for large scale IOT devices.
 - f Explain the five critical requirements for sensor commons project.
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Q4. Attempt any three of the following:

15

- a) What is univariate data analysis? Write objectives of univariate data analysis.
- b) Explain how central tendency helps in univariate data analysis?
- c) What are quartiles? How do they help in plotting a box plot?
- d) What is the difference between a bar chart and histogram? How they help in univariate data analysis.
- e) Highlight the role of skewness in univariate data analysis.
- f) Explain Lorenz curve with an example.

Q5. Attempt any three of the following:

15

- a) Explain Bravais-Pearson Correlation Coefficient with example.
- b) Explain contingency tables with the help of an example.
- c) What is a scatter plot? Explain its significance in understanding the relationship between two variables.
- d) Discuss how to measure the association between mixed combinations of numerical, ordinal, and nominal variables.
- e) What is the Phi Coefficient? Explain its use in measuring association between two binary nominal variables.
- f) Explain how can we measure the association between two variables with different scales?

(2½ Hours)

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 - (2) Answers to the same question must be written together.
 - (3) Numbers to the right indicate marks.
 - (4) Draw neat labeled diagrams wherever necessary.
 - (5) Use of Non-programmable calculators is allowed.

Q1. Attempt any three of the following:

15

- a) Explain different types of data based on measurement.
- b) Differentiate between Structured, Semi-Structured, and Unstructured data with examples.
- c) Define data. Explain the importance of data in exploratory data analysis.
- d) Explain the difference between small and large samples. Discuss the implications of using small vs. large samples in business research.
- e) Define business statistics. Write applications of statistics in business.
- f) Explain the concept of quantitative data and explain how it differs from qualitative data.

Q2. Attempt any three of the following:

15

- a) Write difference between Exploratory Data Analysis and Classical Data Analysis.
- b) Define EDA. Discuss the goals of Exploratory Data Analysis.
- c) Discuss the role of graphics in Exploratory Data Analysis.
- d) Explain Lag plot in detail.
- e) Highlight the significance of Exploratory Data Analysis.
- f) Explain in brief the concepts of unidimensional, bidimensional, graphical representations in EDA.

Q3. Attempt any three of the following:

15

- a) Explain the process of data exploration. Explain how it is helpful for data preparation and analysis?
- b) What is data consistency? Explain why we must ensure consistency of data.
- c) Discuss the problems caused by duplicate or redundant variables in a dataset.
- d) Discuss the issue of missing values in a dataset. Why is handling missing data important during data exploration?
- e) What are Outliers? How are they significant?
- f) How can non-numerical data in missing places be handled during Exploratory Data Analysis (EDA)?

Total Marks: 75

1. Attempt any three of the following:

- 15

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- 15

- 15

- 15

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- a. Define generalization in Machine Learning. Why is it important for building effective models?
- b. Describe the general architecture of Machine Learning systems. What are the key components involved?
- c. Explain the concepts of bias and variance in Machine Learning. How do they affect model performance?
- d. Describe Reinforcement Learning. How does it work, and what are some real-world applications?
- e. What types of problems are best suited for Machine Learning? Explain with examples.
- f. Compare parametric and non-parametric algorithms. Provide examples of each and explain their advantages and disadvantages.

2. Attempt any three of the following:**15**

- a. How does multi-class classification differ from binary classification? Give an example of a multi-class classification problem.
- b. Explain the working of the KNN algorithm. How does it classify new data points?
- c. Describe the working of Decision Trees in Machine Learning. How do they make predictions?
- d. What is classification in Machine Learning? Explain its importance with an example.
- e. How does the value of 'K' impact the performance of the KNN algorithm? What factors should be considered while selecting the optimal 'K' value?
- f. Define entropy and Gini impurity. How are they used in attribute selection for Decision Trees?

3. Attempt any three of the following:**15**

- a. Differentiate between Hard Margin and Soft Margin in SVM. When is each approach used?
- b. Explain the working of the Naïve Bayes Classifier. Why is it called 'Naïve'?
- c. Describe Logistic Regression. How does it differ from Linear Regression?
- d. Explain the concept of Linear Regression. How is it used for predictive modeling?
- e. Interpret Bayes' Rule in the context of Machine Learning. How is it used for probability estimation?
- f. Explain the working of Support Vector Machines (SVM) in classification problems. How does it separate data points?

4. Attempt any three of the following:

15

- a. What is cross-validation in Machine Learning? How does it improve model performance and reliability?
- b. Describe the ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve). How do they help in evaluating classification models?
- c. List and describe at least three real-world applications of Clustering. How does it help in pattern recognition?
- d. Explain two main types of Hierarchical Clustering. How do they work?
- e. Define accuracy in classification. Why might accuracy not always be a reliable metric?
- f. What is the F1 Score? How is it calculated, and why is it useful in imbalanced datasets?

5. Attempt any three of the following:

15

- a. List and describe the main steps involved in the K-Means Clustering algorithm.
- b. What are the main criteria for dimensionality reduction? Why is it important in Machine Learning?
- c. Explain the working of Principal Component Analysis (PCA). How does it help in reducing dimensionality?
- d. What is Market Basket Analysis? How is it used in data mining and business analytics?
- e. What are the main steps involved in the Apriori algorithm? Explain with an example.
- f. Discuss the strengths and weaknesses of K-Means Clustering. In which situations is it most effective?